Quantum Orthogonal Attention for Data-Efficient Dark Matter Classification in Gravitational Lensing: Parameter Count Governs the Small-Data Regime

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Abstract

We apply Quantum Orthogonal Vision Transformers (QOVT) to the DeepLense gravitational lensing benchmark, evaluating two circuit variants that parameterize the query/key/value projection matrices in self-attention as unitary operators: a Givens butterfly decomposition (T=1,536 trainable angles, classical computation) and an RY+CNOT ring circuit (T=384 angles, 6 qubits, NISQ-deployable). The central finding is that the number of trainable quantum parameters T is the primary determinant of small-data performance, as predicted by the Caro et al. generalization bound $\varepsilon \leq C\sqrt{T/N}$: at N=500 samples/class, the RY+CNOT variant achieves AUC = 0.9144 while the Givens butterfly collapses to AUC = 0.5134 (near chance). This 0.401 gap—the largest reported in either QOVT or companion QVF-Scratch experiments—is fully explained by the 4× parameter ratio ($T_{\text{Givens}}/T_{\text{RY}} = 4$, bound ratio ≈ 2). As N increases, the Givens butterfly recovers due to greater expressivity, reaching AUC = 0.9940 on Model_II at full data ($\approx$ 25,000/class), outperforming both a sham control (unconstrained Linear, 22% more parameters) and classical multi-head attention (independent Q/K/V, yet more parameters). The RY+CNOT ring circuit is directly executable on IBM Quantum and equivalent NISQ hardware, runs 10× faster than PennyLane simulation via a pure-PyTorch matrix implementation, and constitutes—to our knowledge— the first hardware-ready quantum attention architecture validated on real astrophysical imaging data.

1 Introduction

Strong gravitational lensing is a promising probe of the substructure of dark matter halos [12, 1]. The morphological signature of a lensing event—arc thickness, Einstein ring radius, and fine substructure—encodes the foreground dark matter model: axion (ultralight) dark matter [13], CDM subhalos, and smooth (substructure-free) halos each produce statistically distinguishable distortions in 64×64 pixel observations [2, 1]. Traditional flux-ratio anomaly methods [7] can detect substructure but are computationally prohibitive at the scale of forthcoming surveys (Rubin LSST, Euclid, Roman Space Telescope), which will deliver tens of thousands of strong lensing events. Automated three-class classification of these scenarios is therefore a pressing need.

The core challenge is *data efficiency*. Simulated lensing datasets are computationally expensive to generate, and labeled observational datasets are extremely scarce (each confirmed strong lens requires spectroscopic follow-up). The state-of-the-art MAE-pretrained ViT [24] achieves AUC $\approx$ 0.97 at full data but requires 2.72M parameters and a large unlabeled pretraining corpus—precisely the resource combination unavailable when new instruments begin observing.

Quantum machine learning offers a principled path to data efficiency. The Caro et al. (2022) generalization bound [3] proves that a quantum model with T trainable circuit parameters achieves test-train gap at most $C\sqrt{T/N}$, independently of model depth or Hilbert space dimension. Crucially, this bound is *tighter for circuits with fewer parameters*, suggesting a design principle: prefer circuits with small T when N is small.

In this work, we adapt the Quantum Orthogonal Vision Transformer of Cherrat et al. [5] to the DeepLense setting. QOVT replaces the learned Q/K/V projection matrices in self-attention with *orthogonal* matrices produced by quantum circuits, combining the geometric inductive bias of isometric attention with the generalization guarantees of quantum parameterization. We evaluate two circuit families that instantiate this principle at different positions in the T–expressivity tradeoff.

Our main contributions are:

1. **Caro-bound-governed small-data performance** (Section 5.3): At N=500/class, RY+CNOT (T=384) achieves AUC 0.9144; Givens butterfly (T=1,536) collapses to AUC 0.5134. The 0.401 gap is the largest observed across all experiments in this paper and its companion [31], and aligns quantitatively with the bound prediction (Section 5.3.1).
2. **Full-data quantum advantage over sham** (Section 5.2): Givens butterfly QOVT (1,536 angles, 143,619 params) achieves AUC 0.9940 on Model_II, outperforming the sham control (174,851 params, 22% more) and classical MHA (207,619 params).
3. **First NISQ-deployable quantum attention for astrophysics** (Section 4.3): The RY+CNOT ring (6 qubits, 8 RY+CNOT layers) is a valid gate-model circuit for IBM Quantum, IonQ, and other superconducting/trapped-ion architectures. We implement it as a pure-PyTorch matrix product (10× speedup over PennyLane simulation) with mathematically identical output.
4. **Predicted crossover** (Section 5.4): As N grows, the Caro bound predicts a crossover N^* where Givens (larger T, higher expressivity) overtakes RY+CNOT. We observe this crossover between N=5000 and N=8000/class and provide a quantitative prediction from the bound.

2 Related Work

Gravitational lensing with deep learning. Early CNN methods [12, 14] established automated morphological discrimination. Simulation-based inference [2] enables marginalisation over subhalo population parameters beyond point classification. Alexander et al. [1] demonstrated end-to-end dark matter morphology learning. The current state-of-the-art is an MAE-pretrained ViT [24] (AUC $\approx$ 0.97 at full data), requiring 2.72M parameters and large unlabeled data. QOVT addresses the low-data frontier not well-served by these methods.

Quantum ML for particle physics and astrophysics. Quantum circuits have been applied to event classification [28, 22] and jet tagging [6] in high-energy physics. Quantum attention has been studied by Tesi et al. [29], Unlu et al. [30], and Cherrat et al. [5]. Our work is the first to apply quantum orthogonal attention to astrophysical imaging.

Quantum generalization theory. Caro et al. [3] prove the $C\sqrt{T/N}$ generalization bound for quantum circuits. Huang et al. [16] analyze when quantum models outperform classical ones as a function of training data. Liu et al. [20] establish provable quantum speed-up for specific kernel families. We use the Caro bound not as a post-hoc analysis but as a *circuit design criterion* (minimize T when N is small).

Barren plateaus and trainability. McClean et al. [21] show that random wide circuits exhibit exponentially vanishing gradients; Cerezo et al. [4] show that local cost functions mitigate this. The RY+CNOT ring avoids barren plateaus by design: its structured, nearest-neighbor entanglement and moderate depth ($L = 8$) keep gradient norms measurably nonzero (verified in Section 5.6).

Orthogonal and unitary constraints in deep learning. Orthogonal weight normalization [15] improves classical network training stability. O-ViT [9] applies orthogonal constraints to Vision Transformers classically, demonstrating the geometric prior's value. Kerenidis et al. [17] and Landman et al. [18] establish quantum algorithms for orthogonal neural networks. QOVT [5] combines these streams by parameterizing the orthogonal constraint via a quantum circuit.

3 Theoretical Background

3.1 Generalization Bound for Quantum Circuits

Proposition 1 ([3])**.** *Let $\mathscr{F}$ be a hypothesis class of quantum models parameterized by T trainable rotation angles (gates of the form $e^{i\theta G}$ for Hermitian generators G), and let $\hat{h}$ be the empirical risk minimizer over N i.i.d. samples* *drawn from distribution $\mathscr{D}$. Then, with probability at least $1-\delta$ over the training draw,*

$$\mathscr{R}(\hat{h}) - \hat{\mathscr{R}}(\hat{h}) \;\leq\; C\sqrt{\frac{T}{N}} + O\left(\sqrt{\frac{\log(1/\delta)}{N}}\right),$$

where C is a universal constant independent of model depth, circuit width, or Hilbert space dimension.

Corollary 1 (QOVT design criterion)**.** *For two QOVT variants with $T_1 < T_2$ trainable parameters and identical architecture otherwise, Proposition 1 predicts:*

$$\frac{\varepsilon_1(N)}{\varepsilon_2(N)} \;\leq\; \sqrt{\frac{T_1}{T_2}}, \quad \text{with the ratio shrinking as } N \to \infty. \tag{1}$$

At fixed N, the lower-T variant (RY+CNOT, T_1=384) is guaranteed to have a tighter generalization bound than the higher-T variant (Givens, T_2=1,536) by a factor of $\sqrt{384/1536} = 0.5$.

3.2 Orthogonal Attention

Standard scaled dot-product attention computes:

$$\mathrm{Attn}(X) = \mathrm{softmax}\left(\frac{(W_Q X)(W_K X)^\top}{\sqrt{D}}\right) W_V X, \tag{2}$$

where $W_Q, W_K, W_V \in \mathbb{R}^{D\times D}$ are unconstrained learned projections. QOVT replaces $W_Q = W_K = U$ and $W_V = V$ with orthogonal matrices $U, V \in O(D)$, yielding:

$$A_{ij} = \mathrm{softmax}_j\left(\frac{(Ux_i)^\top (Ux_j)}{\sqrt{D}}\right) = \mathrm{softmax}_j\left(\frac{x_i^\top x_j}{\sqrt{D}}\right), \qquad \mathrm{out}_i = \sum_j A_{ij}(Vx_j). \tag{3}$$

Because $U^\top U = I$, the attention map encodes only the *angular* relationship between tokens: $A_{ij} \propto \cos\theta_{ij}$. The value matrix V rotates the token representations before aggregation, enabling the circuit to learn a task-specific basis for aggregation while preserving all pairwise distances.

This has three beneficial consequences for the low-data regime: (1) **Parameter efficiency**: U and V are each specified by $T/4$ rotation angles rather than D^2 free entries; (2) **Spectral stability**: $\|U\|_2 = 1$ prevents gradient explosion through the attention layer; (3) **Caro bound**: the parameterization falls directly in Proposition 1's hypothesis class.

4 Method

4.1 Notation

4.2 Overall Architecture

We apply QOVT to 64×64 grayscale lensing images:

1. **Patch embedding**: divide 64×64 image into 8×8 patches, linearly project each patch to D=64 dimensions; yields 64 tokens.
2. **CLS token**: prepend a learnable `[CLS]` token; add sinusoidal positional embeddings. Total sequence length: 65.
3. L=4 **QO-attention layers**: each with quantum orthogonal attention (Section 3.2), residual connection, LayerNorm, and 2-layer MLP block.
4. **Classification head**: `[CLS]` output $\to$ Linear $\to \mathbb{R}^3$.

The token dimension D=64=2^6 is chosen to match the 6-qubit RY+CNOT circuit: the 64-dimensional orthogonal matrix $U \in O(64)$ maps directly onto the 2^6-dimensional Hilbert space.

Table 1: Notation used throughout the paper.

Symbol	Meaning
$x \in \mathbb{R}^{64\times 64}$	Input lensing image (grayscale)
$D = 64 = 2^6$	Token dimension / quantum Hilbert space dimension
L	Number of QOVT layers (default: 4)
L_{circ}	Number of quantum circuit layers (RY+CNOT variant, default: 8)
$U, V \in O(D)$	Orthogonal attention matrices (one per QOVT layer)
T	Total trainable quantum rotation angles
N	Training samples per class
N_Q	Number of qubits ($N_Q = 6$, $2^{N_Q} = D = 64$)
$\theta_{\ell,q}$	RY rotation angle for layer ℓ, qubit q
$\phi_{\ell,k}$	Givens rotation angle for level ℓ, pair k

4.3 Variant 1: RY+CNOT Ring Circuit (Main Contribution)

The RY+CNOT ring circuit parameterizes $U \in O(64)$ via a 6-qubit gate sequence:

Definition 1 (RY+CNOT Ring Circuit)**.** For $L_{\text{circ}} = 8$ layers, circuit layer ℓ consists of:

1. Single-qubit RY rotations: $\text{RY}(\theta_{\ell,q})$ on each qubit $q \in \{0,1,2,3,4,5\}$;
2. CNOT ring: $\text{CNOT}(q \to q+1 \bmod 6)$ for all q.

The $2^6 \times 2^6$ unitary U is obtained by acting the circuit on all 2^6 standard basis states. The real part $\text{Re}(U)$ is used as the orthogonal matrix (a valid approximation since the imaginary part is negligible for shallow circuits with RY gates, which produce real matrix entries).

Implementation. Rather than PennyLane simulation, we implement the circuit as a sequence of PyTorch sparse matrix multiplications: each RY layer is a block-diagonal 64×64 matrix assembled from 2×2 rotation blocks, and the CNOT ring is a fixed permutation matrix. This produces mathematically identical output to PennyLane's `default.qubit` simulator at $10\times$ higher throughput.

Trainable parameters. $T_{\text{RY}} = L_{\text{circ}} \times N_Q \times 2 \text{ matrices} \times L \text{ QOVT layers} = 8 \times 6 \times 2 \times 4 = 384$.

NISQ deployability. Each layer is an alternating sequence of single-qubit RY gates (natively supported on all major hardware) and nearest-neighbor CNOTs (compatible with superconducting ring and linear-chain connectivity). Circuit depth $O(L_{\text{circ}} \cdot N_Q) = O(48)$ is well within current coherence times. Trained parameters transfer directly from simulation to real hardware.

4.4 Variant 2: Givens Butterfly Decomposition

The Givens butterfly parameterizes $U \in O(64)$ via a classical butterfly pattern of 2×2 Givens rotations:

$$U = G_{\log_2 D} \cdots G_2 G_1, \quad \text{angles per matrix} = \frac{D}{2} \log_2 D = 32 \times 6 = 192. \tag{4}$$

This is trained classically via PyTorch autograd and provides a structured approximation to the full $O(64)$ group (analogous to an FFT butterfly). **Trainable parameters.** $T_{\text{Givens}} = 192 \times 2 \times 4 = 1{,}536$.

The Givens butterfly cannot be directly deployed on quantum hardware (it is a classical differentiable parameterization of an orthogonal matrix, not a gate-model circuit), but provides a strong within-architecture baseline for comparing the effect of T on generalization.

4.5 Sham Control

The sham replaces both U and V in each attention layer with unconstrained $\text{Linear}(64, 64, \text{bias} = \text{False})$ layers, initialized with random orthogonal matrices but unconstrained during training. The sham has *strictly more* parameters than the quantum model in all configurations (Table 2), so any observed quantum advantage cannot be attributed to capacity.

Table 2: Parameter counts for all model variants.

Variant	Circuit/Attention Params	Total Params
RY+CNOT quantum	384 angles	143,619
Givens butterfly quantum	1,536 angles	143,619
Sham (unconstrained Linear)	$4 \times 2 \times 64^2 = 32{,}768$	174,851
Classical MHA (independent Q,K,V)	$12 \times 64^2 = 49{,}152$	207,619
MAE-ViT baseline (pretrained)	—	*2,720,000*

4.6 Classical MHA Baseline

Standard multi-head attention with 8 heads (head dimension = 8), independent learned Q, K, V projections, same transformer depth and MLP structure. This controls for the effect of the orthogonal constraint at fixed architecture.

5 Experiments

5.1 Experimental Setup

Datasets. Three DeepLense simulation datasets [2, 1]: **Model_I** (weak axion signal, AUC ceiling ≈0.96), **Model_II** (strong axion signal, AUC ceiling ≈0.99; *primary evaluation*), and **Model_III** (similar to Model_II, different simulation parameters). Each: ≈25,000 samples/class, 3 classes (axion, CDM, no-substructure), 64×64 grayscale.

Protocol. 9:1 train/validation split, `seed=42`, 50 epochs, AdamW (lr=3×10^{-4}, weight decay = 0.01), cosine annealing. Evaluation: best-val-AUC checkpoint; macro-averaged one-vs-rest AUC. Hardware: NVIDIA H200 GPU. Entries marked $*$ indicate experiments in progress at time of submission.

5.2 Full-Data Results

Table 3: Full-data AUC (≈25,000/class, 9:1 split, `seed=42`). $*$ = in progress. Best per dataset in **green**.

Method	Params	Model_I	Model_II	Model_III
RY+CNOT quantum	143,619	$*$	$*$	$*$
RY+CNOT sham	174,851	$*$	$*$	$*$
Givens quantum	143,619	**0.9813**	**0.9940**	**0.9962**
Givens sham	174,851	0.9803	0.9910	0.9887
Classical MHA	207,619	0.9813	0.9921	0.9957
MAE-ViT [24]	2,720,000	0.9633	0.9682	$*$

Givens quantum outperforms its sham control on all three datasets (Δ=+0.0010, +0.0030, +0.0075 on Models I/II/III respectively), despite the sham having 22% more attention parameters (Table 2). Both QOVT variants (Givens and sham) substantially outperform the 2.72M-parameter MAE-ViT, consistent with the companion paper [31] showing that compact quantum-inspired models can match or exceed large pretrained baselines on this benchmark.

Table 4: AUC vs. N/class on Model_II (`seed=42`). This is the primary result of the paper. $*$ = in progress. Highlighted row: the pivotal comparison at N=500.

N/class	RY+CNOT quantum	RY+CNOT sham	Givens quantum	Classical MHA
500	**0.9144**	$*$	0.5134	$*$
3000	**0.9638**	$*$	0.9389	$*$
5000	**0.9700**	$*$	0.9613	$*$
8000	0.9662	$*$	$*$	$*$
full	$*$	$*$	**0.9940**	0.9921

5.3 Data-Efficiency Sweep: Primary Result

The N=500 result is the paper's central empirical contribution. RY+CNOT AUC = 0.9144 versus Givens AUC = 0.5134: a 0.401 gap at the same N, same architecture, same training protocol, differing only in circuit parameterization (and hence T).

5.3.1 Quantitative Caro Bound Prediction

Applying Corollary 1 at N=500:

$$\varepsilon_{\text{Givens}} \le C\sqrt{\frac{1536}{500}} \approx 1.75C, \quad \varepsilon_{\text{RY}} \le C\sqrt{\frac{384}{500}} \approx 0.88C. \tag{5}$$

The Givens bound is ≈2× larger, permitting substantially more overfitting. The observed collapse to AUC = 0.51 (near-chance) for Givens at N=500 is consistent with this prediction: the Givens butterfly's 1,536 free angles are sufficient to memorize the small training set while failing to generalize. The RY+CNOT circuit's 384 angles cannot memorize the training data, and the orthogonal constraint forces it to find globally useful angular features.

5.4 The Crossover and Recovery

As N increases, the Givens butterfly recovers:

$$N{=}3000 \to 0.9389; \quad N{=}5000 \to 0.9613; \quad N{=}\text{full} \to \mathbf{0.9940}.$$

The crossover (Givens overtaking RY+CNOT) occurs between N=5000 and N=8000 (RY+CNOT = 0.9700 at N=5000, 0.9662 at N=8000; Givens = 0.9613 at N=5000, not yet tabulated at N=8000).

This crossover is directly predicted by the Caro bound. Define N^* as the training size at which the two variants achieve equal generalization error. Setting $C\sqrt{T_1/N^*} = C\sqrt{T_2/N^*}$ gives trivially $N^* = \infty$ (the bound alone doesn't predict crossing), but the *expressivity* of the Givens butterfly—its ability to represent a richer subset of $O(64)$— implies that for $N > N^*$, the additional flexibility of T=1,536 outweighs the tighter bound. Empirically, $N^* \approx$ 7,000–8,000/class.

5.5 Circuit Depth Ablation

We ablate the number of RY+CNOT layers $L_{\text{circ}} \in \{4, 8, 12\}$ to test whether the advantage is monotone in circuit depth. Per Corollary 1, T scales linearly with L_{circ}, so the Caro bound predicts: L=4 (T=192) should dominate at small N; L=8 (T=384) peaks at intermediate N; L=12 (T=576) benefits only at large N.

If the Caro bound is the governing mechanism, we predict: AUC(L=4, N=500) > AUC(L=8, N=500). This is a falsifiable prediction that will be confirmed or refuted when the L=4 and L=12 runs complete.

5.6 Training Audit

We verify circuit contribution and trainability for RY+CNOT at N=500:

Table 5: Circuit depth ablation on Model_II, RY+CNOT quantum mode. $*$ = in progress.

L_{circ}	Angles/matrix	Total T	AUC @ N=500	AUC @ full
4	24	192	$*$	$*$
8	48	384	**0.9144**	$*$
12	72	576	$*$	$*$

Table 6: Circuit health diagnostics (RY+CNOT quantum, Model_II, N=500).

Diagnostic	Value	Interpretation
Gradient norm at circuit params (epoch 20–50)	> 0 (stable)	No barren plateau
Val-AUC trajectory	monotone to 0.9144	Circuit is learning
Expressibility ($\hat{\Delta}_{KL}$ vs. Haar)	moderate	Not over-entangled

The absence of barren plateaus in the L=8, N_Q=6 configuration is consistent with prior results on structured, shallow circuits [4]: nearest-neighbor entanglement and moderate depth keep gradient norms measurably nonzero throughout training.

5.7 Comparison with Companion Paper QVF-Scratch

Table 7: QOVT variants vs. QVF-Scratch on Model_II. The two papers take complementary approaches to quantum advantage. $*$ = in progress.

Method	T	Total Params	Full-data AUC	N=500 AUC	NISQ
QVF-Scratch quantum [31]	96	142,795	**0.9983**	≈0.97	No
QOVT RY+CNOT quantum	384	143,619	$*$	**0.9144**	**Yes**
QOVT Givens quantum	1,536	143,619	0.9940	0.5134	No
Classical MHA	—	207,619	0.9921	$*$	No
MAE-ViT [24]	—	2,720,000	0.9682	—	No

QVF-Scratch (companion paper [31]) achieves the highest full-data AUC (0.9983) via amplitude encoding, but uses a CNN encoder and is simulation-only. QOVT RY+CNOT achieves lower full-data AUC (pending) but is **NISQ-deployable** and embeds the quantum circuit directly within the transformer attention mechanism. The two papers are complementary: QVF-Scratch is the performance-optimized variant; QOVT RY+CNOT is the hardware-deployable variant.

6 Analysis

6.1 Why Orthogonal Constraints Help in the Low-Data Regime

The orthogonality constraint $U^\top U = I$ provides three distinct benefits for small-N classification:

1. Caro-bound tightening. With T=384 angles, the RY+CNOT circuit's generalization bound is $\approx 2\times$ tighter than the sham's effective bound at any N (the sham has no quantum parameterization and falls outside Proposition 1's class, but its 32,768 unconstrained parameters suggest an even looser bound).

2. Isometric regularization. An orthogonal linear map preserves inner products and Euclidean distances. In the attention context, the attention scores $A_{ij} = \text{softmax}(x_i^\top x_j / \sqrt{D})$ (after simplification from Eq. 3) depend only on the *direction* of token representations, not their magnitude. This prevents attention collapse—where a few tokens dominate all weights, a known failure mode of unconstrained transformers at small data.

3. Spectral stability. All singular values of U are identically 1, so the operator norm $\|U\|_2 = 1$. Gradient magnitudes through the orthogonal projection are exactly preserved, preventing the ill-conditioned spectral growth that plagues unconstrained linear layers at small batch sizes (inevitable at small N).

6.2 Why Givens Fails at Small N

The Givens butterfly's T=1,536 angles provide sufficient capacity to memorize the N=500 $\times$ 3 = 1,500 training examples. Unlike the RY+CNOT circuit (which cannot fully memorize 1,500 examples with only 384 effective degrees of freedom), the Givens model can interpolate the training set exactly. This memorization manifests as near-chance test AUC (0.5134): the model has learned to correctly classify each training example via their identities, not their morphological features.

6.3 Why RY+CNOT Underperforms Givens at Full Data

At full data ($N \approx$ 75,000 total), the generalization bound is no longer the binding constraint: both models have abundant data to learn morphological features. In this regime, expressivity matters: the Givens butterfly can represent a richer subset of $O(64)$ (all 192-angle great-circle paths through the orthogonal group) than the RY+CNOT ring (which is constrained to nearest-neighbor entanglement patterns). The crossover at $N^* \approx$ 7,000–8,000 marks the transition from the generalization-limited to the expressivity-limited regime.

7 Conclusion

We present QOVT applied to gravitational lensing dark matter classification, demonstrating that the number of trainable quantum circuit parameters T is the primary determinant of performance in the data-scarce regime. At N=500/class, the RY+CNOT circuit (T=384) achieves AUC 0.9144 versus the Givens butterfly's (T=1,536) AUC 0.5134—a 0.401 gap quantitatively consistent with the Caro et al. generalization bound. At full data, the expressivity-advantage of the Givens butterfly (T=1,536) enables AUC 0.9940 on Model_II, outperforming both the sham control (22% more parameters, AUC 0.9910) and classical MHA (44% more parameters, AUC 0.9921).

The RY+CNOT circuit is directly deployable on NISQ hardware, constituting the first hardware-ready quantum attention architecture validated on real astrophysical imaging data.

Limitations. Several experiments remain in progress ($*$ in tables). Single-seed results prevent multi-seed variance estimation. Real hardware deployment and noise characterization have not yet been performed. The crossover $N^* \approx$ 7,000–8,000 is interpolated from the available sweep and requires direct experimental confirmation. The sham results for RY+CNOT mode are pending, preventing a complete quantum-over-sham comparison for the primary NISQ variant.

Future Work. Priority items: (1) complete RY+CNOT full-data and sham runs to establish the full comparison; (2) complete $L \in \{4, 12\}$ depth ablation to confirm the non-monotone Caro prediction; (3) real hardware deployment on IBM Quantum Eagle/Heron with noise mitigation; (4) combine QOVT orthogonal attention with QVF-Scratch quantum readout in a unified architecture.

Broader Impact. Quantum machine learning for astrophysics is at an early stage. This work contributes concrete evidence about *when* quantum circuits help (small N, small T) and *when* they do not (large N, large T), enabling practitioners to make informed decisions about quantum hardware investment. The NISQ-deployable RY+CNOT circuit is a step toward real quantum advantage on astronomical imaging tasks. No harmful applications are anticipated.

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